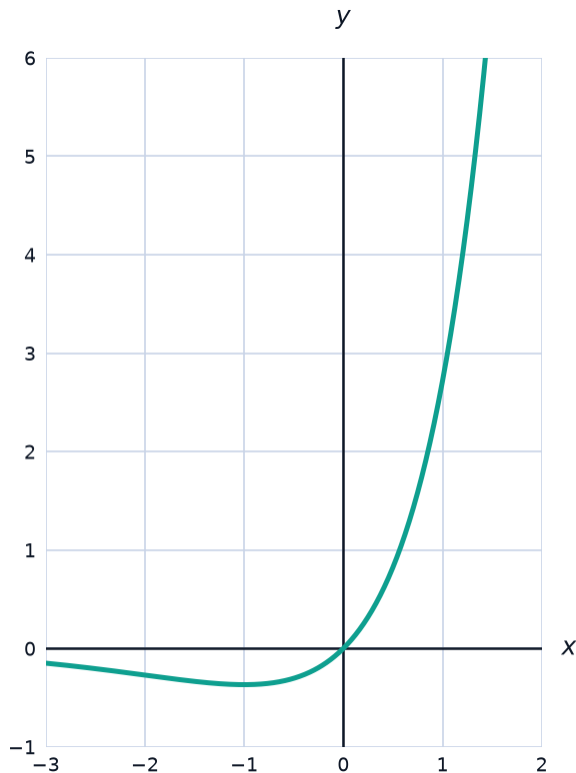


Integration by Parts



The integration by parts formula

- 1 State the integration by parts formula, and explain the LIATE guideline for choosing u .
- 2 Derive the integration by parts formula starting from the product rule $\frac{d}{dx}[uv] = u'v + uv'$.
- 3 Evaluate $\int xe^x dx$.



- 4 Evaluate $\int x \cos x dx$.

Integration by parts with logarithms and inverse trig

- 5 Evaluate $\int \ln x dx$.
- 6 Evaluate $\int x \ln x dx$.
- 7 Evaluate $\int \tan^{-1} x dx$.
- 8 Evaluate $\int x^2 \ln x dx$.

- 9 Evaluate $\int \sin^{-1} x \, dx$.
-

Repeated integration by parts

- 10 Evaluate $\int x^2 e^x \, dx$, applying integration by parts twice.
- 11 Evaluate $\int x^2 \sin x \, dx$, applying integration by parts twice.
- 12 Set up (do not evaluate fully) the reduction needed to find $\int x^3 e^x \, dx$, identifying how many times integration by parts must be applied.
-

The cyclic case

- 13 Evaluate $\int e^x \sin x \, dx$ using the cyclic (solve-for-the-integral) technique.
- 14 Evaluate $\int e^{2x} \cos x \, dx$ using the cyclic technique.
-

Definite integrals using integration by parts

- 15 Evaluate $\int_0^1 x e^x \, dx$ using the antiderivative found earlier.
- 16 Evaluate $\int_1^e \ln x \, dx$ using the antiderivative found earlier.
- 17 Evaluate $\int_0^\pi x \sin x \, dx$.
-

Integration by parts to derive a reduction formula

- 18 Use integration by parts to show $\int x^n e^x \, dx = x^n e^x - n \int x^{n-1} e^x \, dx$ (the reduction formula), and use it once to reconfirm $\int x^2 e^x \, dx$.
-

Choosing u and dv wisely

- 19 For each integral, state the best choice of u (using LIATE) without evaluating:
- | | |
|------------------------------|---------------------------|
| a $\int x^3 \ln x \, dx$ | b $\int x e^{5x} \, dx$ |
| c $\int x \tan^{-1} x \, dx$ | d $\int x^2 \cos x \, dx$ |

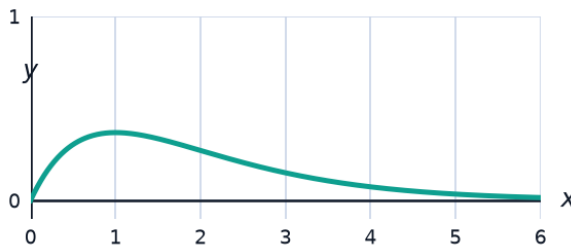
- 20 Explain what goes wrong if you attempt $\int x e^x dx$ with the reversed choice $u = e^x$, $dv = x dx$.
-

Integration by parts with algebraic-trigonometric products

- 21 Evaluate $\int x \sec^2 x dx$.
- 22 Evaluate $\int 2x \sin(3x) dx$.
-

Definite integrals: area under a transcendental product

- 23 Find the exact area under $y = x e^{-x}$ on $[0, 3]$, using $\int x e^{-x} dx = -x e^{-x} - e^{-x} + C$ (verify this antiderivative first by differentiating).



- 24 Evaluate $\int_0^{\pi/2} x \cos x dx$ using the antiderivative $x \sin x + \cos x$ found earlier.
-

Mixed review: choosing the right technique

- 25 For each integral, state whether it should be solved by u -substitution or integration by parts, and identify the key choice (substitution variable or u) without fully evaluating:
- | | |
|-----------------------------|---------------------|
| a $\int 2x e^{x^2} dx$ | b $\int x^2 e^x dx$ |
| c $\int \frac{\ln x}{x} dx$ | d $\int x \ln x dx$ |
-

Multiple choice — easy 15

26 Evaluate $\int_0^{\pi} -2x \cos x \, dx$, using integration by parts.

- a. 2
- b. 6
- c. 4
- d. -4

27 Evaluate $\int_0^{\pi} 3x \cos x \, dx$, using integration by parts.

- a. -6
- b. -3
- c. 6
- d. -4

28 Evaluate $\int_0^{\pi} -6x \cos x \, dx$, using integration by parts.

- a. 6
- b. 14
- c. 12
- d. -12

29 Evaluate $\int_0^{\pi} x \cos x \, dx$, using integration by parts.

- a. 2
- b. 0
- c. -1
- d. -2

30 Evaluate $\int_0^{\pi} -7x \cos x \, dx$, using integration by parts.

- a. 16
- b. 7
- c. -14
- d. 14

31 Evaluate $\int_0^{\pi} 7x \cos x \, dx$, using integration by parts.

- a. -14
- b. -12
- c. -7
- d. 14

32 Evaluate $\int_0^{\pi} 9x \cos x \, dx$, using integration by parts.

- a. -16
- b. 18
- c. -18
- d. -9

33 Evaluate $\int_0^{\pi} -8x \cos x \, dx$, using integration by parts.

- a. 18
- b. 16
- c. -16
- d. 8

34 Evaluate $\int_0^{\pi} 8x \cos x \, dx$, using integration by parts.

a. -16

b. 16

c. -14

d. -8

35 Evaluate $\int_0^{\pi} -x \cos x \, dx$, using integration by parts.

a. -2

b. 4

c. 2

d. 1

36 Evaluate $\int_0^{\pi} 6x \cos x \, dx$, using integration by parts.

a. 12

b. -12

c. -10

d. -6

37 Evaluate $\int_0^{\pi} -3x \cos x \, dx$, using integration by parts.

a. 3

b. -6

c. 8

d. 6

38 Evaluate $\int_0^{\pi} -9x \cos x \, dx$, using integration by parts.

- a. 9
- b. -18
- c. 20
- d. 18

39 Evaluate $\int_0^{\pi} -4x \cos x \, dx$, using integration by parts.

- a. 4
- b. -8
- c. 10
- d. 8

40 Evaluate $\int_0^{\pi} 5x \cos x \, dx$, using integration by parts.

- a. -8
- b. 10
- c. -10
- d. -5

Multiple choice — medium 15

41 Given $f(0) = 0$, $f(3) = 2$, $g(0) = 2$, $g(3) = 2$, and $\int_0^3 f'(x)g(x) \, dx = -2$, use integration by parts to find $\int_0^3 f(x)g'(x) \, dx$.

- a. 10
- b. -6
- c. 2
- d. 6

- 42 Given $f(0) = 0$, $f(1) = 2$, $g(0) = 3$, $g(1) = 5$, and $\int_0^1 f'(x)g(x) dx = 7$, use integration by parts to find $\int_0^1 f(x)g'(x) dx$.
- a. 7
 - b. -3
 - c. 17
 - d. 3
- 43 Given $f(2) = 4$, $f(7) = 4$, $g(2) = 0$, $g(7) = -2$, and $\int_2^7 f'(x)g(x) dx = -14$, use integration by parts to find $\int_2^7 f(x)g'(x) dx$.
- a. 10
 - b. -22
 - c. -6
 - d. 6
- 44 Given $f(1) = 5$, $f(3) = 6$, $g(1) = 4$, $g(3) = -6$, and $\int_1^3 f'(x)g(x) dx = -7$, use integration by parts to find $\int_1^3 f(x)g'(x) dx$.
- a. -45
 - b. -49
 - c. -63
 - d. 49
- 45 Given $f(0) = -5$, $f(4) = 6$, $g(0) = -1$, $g(4) = 1$, and $\int_0^4 f'(x)g(x) dx = -12$, use integration by parts to find $\int_0^4 f(x)g'(x) dx$.
- a. -13
 - b. -11
 - c. 13
 - d. 17

- 46 Given $f(3) = -1$, $f(6) = 2$, $g(3) = 1$, $g(6) = 3$, and $\int_3^6 f'(x)g(x) dx = -8$, use integration by parts to find $\int_3^6 f(x)g'(x) dx$.
- a. -1
 - b. 15
 - c. -15
 - d. 19
- 47 Given $f(3) = -2$, $f(6) = 6$, $g(3) = 6$, $g(6) = 3$, and $\int_3^6 f'(x)g(x) dx = -8$, use integration by parts to find $\int_3^6 f(x)g'(x) dx$.
- a. 38
 - b. 42
 - c. -38
 - d. 22
- 48 Given $f(3) = 5$, $f(7) = 0$, $g(3) = 6$, $g(7) = -5$, and $\int_3^7 f'(x)g(x) dx = -10$, use integration by parts to find $\int_3^7 f(x)g'(x) dx$.
- a. -40
 - b. -20
 - c. 20
 - d. -16
- 49 Given $f(0) = -1$, $f(5) = 4$, $g(0) = -2$, $g(5) = -3$, and $\int_0^5 f'(x)g(x) dx = 11$, use integration by parts to find $\int_0^5 f(x)g'(x) dx$.
- a. 25
 - b. -21
 - c. -3
 - d. -25

- 50 Given $f(0) = 1$, $f(4) = 1$, $g(0) = 4$, $g(4) = 0$, and $\int_0^4 f'(x)g(x) dx = -5$, use integration by parts to find $\int_0^4 f(x)g'(x) dx$.
- a. -1
 - b. 1
 - c. -9
 - d. 5
- 51 Given $f(0) = 5$, $f(3) = -2$, $g(0) = 3$, $g(3) = 1$, and $\int_0^3 f'(x)g(x) dx = -6$, use integration by parts to find $\int_0^3 f(x)g'(x) dx$.
- a. -7
 - b. -11
 - c. -23
 - d. 11
- 52 Given $f(1) = -5$, $f(5) = -3$, $g(1) = 3$, $g(5) = -1$, and $\int_1^5 f'(x)g(x) dx = 4$, use integration by parts to find $\int_1^5 f(x)g'(x) dx$.
- a. 18
 - b. 22
 - c. 14
 - d. -14
- 53 Given $f(3) = -4$, $f(4) = 6$, $g(3) = -4$, $g(4) = -6$, and $\int_3^4 f'(x)g(x) dx = -2$, use integration by parts to find $\int_3^4 f(x)g'(x) dx$.
- a. -46
 - b. -50
 - c. -54
 - d. 50

- 54 Given $f(3) = -6$, $f(7) = 5$, $g(3) = 0$, $g(7) = 0$, and $\int_3^7 f'(x)g(x) dx = -8$, use integration by parts to find $\int_3^7 f(x)g'(x) dx$.
- a. 12
 - b. -8
 - c. 8
 - d. 15
- 55 Given $f(0) = 0$, $f(3) = -6$, $g(0) = -3$, $g(3) = 4$, and $\int_0^3 f'(x)g(x) dx = -4$, use integration by parts to find $\int_0^3 f(x)g'(x) dx$.
- a. -28
 - b. 20
 - c. -20
 - d. -16

Multiple choice — hard 15

- 56 Using the cyclic technique, $\int e^x \cos(2x) dx = \frac{e^x(\cos(2x) + C_1 \sin(2x))}{5} + C$. Find the coefficient C_1 .
- a. $\frac{1}{5}$
 - b. $-\frac{2}{5}$
 - c. 2
 - d. $\frac{2}{5}$
- 57 Using the cyclic technique, $\int e^{3x} \cos(3x) dx = \frac{e^{3x}(3\cos(3x) + C_1 \sin(3x))}{18} + C$. Find the coefficient C_1 .
- a. $\frac{1}{3}$
 - b. $-\frac{1}{6}$
 - c. $\frac{1}{6}$
 - d. $\frac{7}{6}$

- 58 Using the cyclic technique, $\int e^{3x} \cos(5x) dx = \frac{e^{3x}(3\cos(5x) + C_1 \sin(5x))}{34} + C$. Find the coefficient C_1 .
- a. $\frac{5}{9}$
 - b. $\frac{5}{34}$
 - c. $-\frac{5}{34}$
 - d. $\frac{3}{34}$
- 59 Using the cyclic technique, $\int e^x \cos(5x) dx = \frac{e^x(\cos(5x) + C_1 \sin(5x))}{26} + C$. Find the coefficient C_1 .
- a. 5
 - b. $\frac{1}{26}$
 - c. $-\frac{5}{26}$
 - d. $\frac{5}{26}$
- 60 Using the cyclic technique, $\int e^{5x} \cos(4x) dx = \frac{e^{5x}(5\cos(4x) + C_1 \sin(4x))}{41} + C$. Find the coefficient C_1 .
- a. $-\frac{4}{41}$
 - b. $\frac{4}{41}$
 - c. $\frac{5}{41}$
 - d. $\frac{4}{25}$
- 61 For $\int x \ln x dx$, which is the correct choice of u according to LIATE?
- a. It does not matter which factor is chosen for u
 - b. $u = x$ (Algebraic always goes first)
 - c. $u = x$ (the opposite LIATE priority)
 - d. $u = \ln x$ (Logarithmic outranks Algebraic in LIATE)

- 62 For $\int x^2 \tan^{-1} x \, dx$, which is the correct choice of u according to LIATE?
- a. $u = \tan^{-1} x$ (Inverse Trig outranks Algebraic in LIATE)
 - b. $u = x^2$ (Algebraic always goes first)
 - c. It does not matter which factor is chosen for u
 - d. $u = x^2$ (the opposite LIATE priority)
- 63 For $\int x^3 \ln x \, dx$, which is the correct choice of u according to LIATE?
- a. It does not matter which factor is chosen for u
 - b. $u = x^3$ (the opposite LIATE priority)
 - c. $u = \ln x$ (Logarithmic outranks Algebraic in LIATE)
 - d. $u = x^3$ (Algebraic always goes first)
- 64 For $\int x^2 e^{6x} \, dx$, which is the correct choice of u according to LIATE?
- a. $u = x^2$ (Algebraic outranks Exponential in LIATE)
 - b. $u = x^2$ (Algebraic always goes first)
 - c. It does not matter which factor is chosen for u
 - d. $u = e^{6x}$ (the opposite LIATE priority)
- 65 For $\int x^3 \sin(6x) \, dx$, which is the correct choice of u according to LIATE?
- a. $u = \sin(6x)$ (the opposite LIATE priority)
 - b. It does not matter which factor is chosen for u
 - c. $u = x^3$ (Algebraic outranks Trigonometric in LIATE)
 - d. $u = x^3$ (Algebraic always goes first)

- 66 Given $f(2) = 5$, $f(5) = 2$, $g(2) = 5$, $g(5) = 2$, and $\int_2^5 f(x)g'(x) dx = -1$, use integration by parts to find $\int_2^5 f'(x)g(x) dx$.
- a. -22
 - b. -20
 - c. -16
 - d. 20
- 67 Given $f(1) = 6$, $f(4) = 5$, $g(1) = 4$, $g(4) = 0$, and $\int_1^4 f(x)g'(x) dx = -14$, use integration by parts to find $\int_1^4 f'(x)g(x) dx$.
- a. -6
 - b. -38
 - c. 10
 - d. -10
- 68 Given $f(1) = 5$, $f(3) = -1$, $g(1) = 2$, $g(3) = 0$, and $\int_1^3 f(x)g'(x) dx = -6$, use integration by parts to find $\int_1^3 f'(x)g(x) dx$.
- a. 0
 - b. 4
 - c. -4
 - d. -16
- 69 Given $f(0) = 2$, $f(2) = -2$, $g(0) = 0$, $g(2) = -3$, and $\int_0^2 f(x)g'(x) dx = -10$, use integration by parts to find $\int_0^2 f'(x)g(x) dx$.
- a. -4
 - b. 16
 - c. 20
 - d. -16

70 Given $f(1) = 0$, $f(4) = 6$, $g(1) = -5$, $g(4) = -6$, and $\int_1^4 f(x)g'(x) dx = 1$, use integration by parts to find $\int_1^4 f'(x)g(x) dx$.

- a. 37
- b. -37
- c. -35
- d. -33